

Dream 7B: Diffusion Large Language Models

Jiacheng Ye^{1*} Zhihui Xie^{1*} Lin Zheng^{1*} Jiahui Gao^{2*} Zirui Wu¹

Xin Jiang² Zhenguo Li² Lingpeng Kong¹

¹The University of Hong Kong ² Huawei Noah's Ark Lab

 DreamLM/Dream  Dream-7B

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Abstract

We introduce Dream 7B, the most powerful open diffusion large language model to date. Unlike autoregressive (AR) models that generate tokens sequentially, Dream 7B employs discrete diffusion modeling to refine sequences in parallel through iterative denoising. Our model consistently outperforms existing diffusion language models on general, mathematical, and coding tasks. Dream 7B demonstrates superior planning abilities and inference flexibility, including arbitrary-order generation, infilling capabilities, and tunable quality-speed trade-offs. These results are achieved through simple yet effective training techniques, including AR-based LLM initialization and context-adaptive token-level noise rescheduling. We release both Dream-Base and Dream-Instruct to facilitate further research in diffusion-based language modeling.

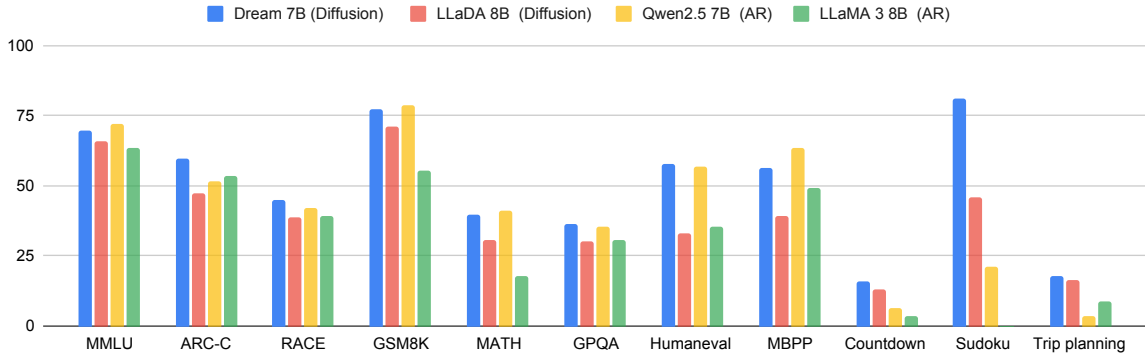


Figure 1: Performance comparison of diffusion and autoregressive language models on various tasks.

1 Introduction

The rapid advancement of large language models (LLMs) has fundamentally transformed artificial intelligence, driving breakthroughs across diverse applications and industries. The current landscape is dominated by autoregressive (AR) models that generate text through sequential left-to-right token prediction, with virtually all leading systems including GPT-4 (OpenAI et al., 2024) and DeepSeek R1 (Guo et al., 2025). While these models have demonstrated remarkable capabilities, a fundamental question emerges: What architectural paradigms might define the next generation of LLMs? This question becomes increasingly important as AR models exhibit inherent limitations in complex reasoning, long-term planning, and maintaining coherence across extended contexts (Bubeck et al., 2023; Dziri et al., 2023; Bachmann and Nagarajan, 2024; Ye et al., 2025a). These limitations are particularly crucial

*Equal Contribution

梦想 7B：扩散大型语言模型

叶家成¹*谢志辉¹*郑林¹*高家辉²*吴自瑞¹蒋欣²李振国²孔令鹏¹ ¹香港大学²华为诺亚方舟实验室DreamLM/Dream Dream-7B



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抽象的

我们介绍 Dream 7B，迄今为止最强大的开放扩散大型语言模型。与顺序生成标记的自回归 (AR) 模型不同，Dream 7B 采用离散扩散模型，通过迭代去噪并行细化序列。我们的模型在一般、数学和编码任务上始终优于现有的扩散语言模型。Dream 7B 展示了卓越的规划能力和推理灵活性，包括任意订单生成、填充功能和可调的质量速度权衡。这些结果是通过简单而有效的训练技术实现的，包括基于 AR 的 LLM 初始化和上下文自适应令牌级噪声重新调度。我们发布了 Dream-Base 和 Dream-Instruct，以促进基于扩散的语言建模的进一步研究。

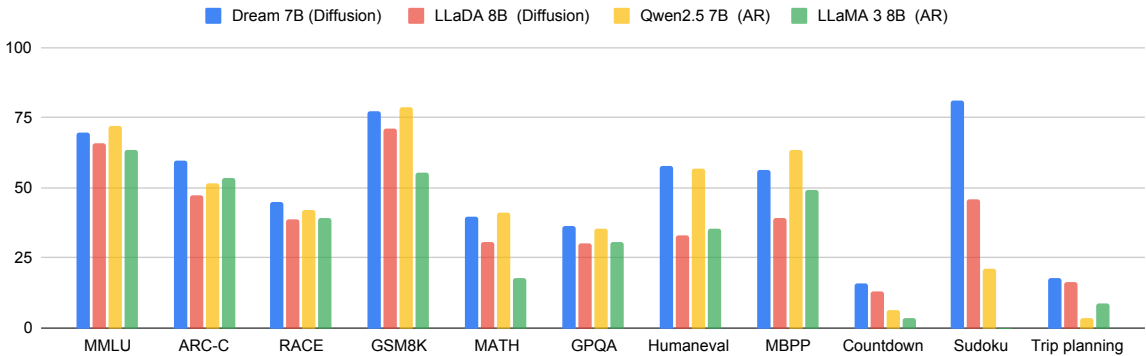


图 1：扩散和自回归语言模型在各种任务上的性能比较。

1 简介

大语言模型 (LLM) 的快速发展从根本上改变了人工智能，推动了不同应用和行业的突破。当前的格局以自回归 (AR) 模型为主，该模型通过顺序从左到右的标记预测生成文本，几乎所有领先的系统都包括 GPT-4 (OpenAI 等人, 2024 年) 和 DeepSeek R1 (Guo 等人, 2025 年)。虽然这些模型展示了卓越的功能，但出现了一个基本问题：什么样的架构范式可以定义下一代法学硕士？由于 AR 模型在复杂推理、长期规划和跨扩展上下文保持一致性方面表现出固有的局限性，这个问题变得越来越重要 (Bubec k 等人, 2023; Dziri 等人, 2023; Bachmann 和 Nagarajan, 2024; Ye 等人, 2025a)。这些限制尤其重要

*Equal Contribution

for emerging applications such as embodied AI, autonomous agents, and long-horizon decision-making systems, where the ability to reason holistically about entire sequences rather than incrementally becomes paramount.

Discrete diffusion models (DMs) have gained attention as a promising alternative for sequence generation since their introduction to the text domain (Austin et al., 2021a; Hooeboom et al., 2021; Campbell et al., 2022). Rather than building text token by token from left to right, diffusion models begin with completely corrupted sequences and gradually denoise them through multiple refinement steps, enabling bidirectional context integration and more flexible generation patterns. This architectural shift unlocks several theoretical advantages that directly address autoregressive limitations. The parallel nature of diffusion enables richer contextual modeling by incorporating information from all positions simultaneously, potentially leading to more coherent and globally consistent outputs (Li et al., 2022). Furthermore, the iterative refinement process naturally supports controllable generation (Gulrajani and Hashimoto, 2023b; Li et al., 2022) and provides flexible quality-speed trade-offs through adjustable inference steps, offering new dimensions for optimization for test-time scaling (Snell et al., 2024; Muennighoff et al., 2025; Geiping et al., 2025) that complement existing techniques like long-thought chain-of-thought reasoning (Jaech et al., 2024; Guo et al., 2025).

Recent developments have begun to demonstrate the practical viability of scaled diffusion language models. Works such as DiffuLLaMA (Gong et al., 2025), LLaDA (Nie et al., 2025b) have successfully scaled discrete diffusion to billion-parameter regimes. While Mercury Coder (Inception Labs, 2025), as a commercial implementation, has demonstrated remarkable inference efficiency in code generation. This rapid progress, combined with the inherent architectural advantages of diffusion language modeling, positions these models as a promising direction for overcoming the fundamental limitations of autoregressive approaches. However, significant gaps remain in achieving performance parity with state-of-the-art autoregressive models like Qwen2.5 (Qwen et al., 2025) across various general tasks (Clark et al., 2018; Hendrycks et al., 2020).

In this work, we present Dream 7B, a 7-billion parameter diffusion language model that bridges this performance gap while demonstrating unique capabilities inherent to the diffusion paradigm. We introduce a comprehensive training framework that leverages AR-based LLM initialization and context-adaptive noise scheduling to scale diffusion language models. Through extensive evaluation, we establish that diffusion models can match AR-based LLM performance on general language tasks while providing distinct advantages in planning-intensive scenarios and offering unprecedented inference flexibility through arbitrary-order generation and dynamic quality-speed optimization. We release both Dream-Base and Dream-Instruct to facilitate further research in this promising direction.

Takeaways

- We release Dream 7B, which consistently outperforms existing diffusion language models by substantial margins across diverse benchmarks.
- Dream 7B achieves competitive performance with Qwen 2.5 on standard benchmarks (general language understanding, mathematical reasoning, and code generation) while exhibiting superior planning abilities and novel inference flexibility features that naturally emerge from the diffusion modeling paradigm.

2 Related Work

2.1 Diffusion Models for Text

Diffusion models were originally developed for continuous domains like image generation (Song et al., 2021; Ho et al., 2020). Building on their remarkable success, a number of works have extended diffusion techniques to text generation tasks by modeling text in a continuous embedding space, building continuous diffusion language models (Li et al., 2022; Gong et al., 2023b,a; Dieleman et al., 2022). Among these approaches, Lin et al. (2023) demonstrated the feasibility of applying diffusion models within a pre-training and fine-tuning framework on a small-scale transformer.

To fit the discrete nature of text, Austin et al. (2021a); Hooeboom et al. (2021); Campbell et al. (2022) introduced discrete diffusion, which operates on the discrete vocabulary space, establishing the

对于嵌入式人工智能、自主代理和长期决策系统等新兴应用来说，对整个序列进行整体推理而不是增量推理的能力变得至关重要。

离散扩散模型（DM）自从被引入文本领域以来，作为一种有前途的序列生成替代方案而受到关注（Austin 等人，2021a；Hoogeboom 等人，2021；Campbell 等人，2022）。扩散模型不是从左到右逐个构建文本标记，而是从完全损坏的序列开始，并通过多个细化步骤逐渐对它们进行去噪，从而实现双向上下文集成和更灵活的生成模式。这种架构转变释放了几个直接解决自回归限制的理论优势。扩散的并行性质可以通过同时合并来自所有位置的信息来实现更丰富的上下文建模，从而可能导致更加连贯和全局一致的输出（Li et al., 2022）。此外，迭代细化过程自然支持可控生成（Gulrajani 和 Hashimoto, 2023b；Li 等人，2022），并通过可调整的推理步骤提供灵活的质量速度权衡，为测试时间扩展的优化提供新的维度（Snell 等人，2024；Muennighoff 等人，2025；Geiping 等人，2025），补充现有技术，例如长期思想链式推理（Jaech et al., 2024；Guo et al., 2025）。

最近的发展已经开始证明缩放扩散语言模型的实际可行性。DiffuLLaMA (Gong et al., 2025)、LLaD A (Nie et al., 2025b) 等工作已成功地将离散扩散扩展到十亿参数范围。而Mercury Coder (Inception Labs, 2025) 作为一种商业实现，在代码生成方面表现出了卓越的推理效率。这种快速的进展，加上扩散语言建模固有的架构优势，使这些模型成为克服自回归方法基本局限性的有希望的方向。然而，在各种一般任务中，与 Qwen2.5 (Qwen 等人，2025) 等最先进的自回归模型实现性能相当方面仍存在显著差距（Clark 等人，2018 年；Hendrycks 等人，2020 年）。

在这项工作中，我们提出了 Dream 7B，这是一个 70 亿参数的扩散语言模型，它弥补了这一性能差距，同时展示了扩散范式固有的独特功能。我们引入了一个综合训练框架，该框架利用基于 AR 的 LLM 初始化和上下文自适应噪声调度来扩展扩散语言模型。通过广泛的评估，我们确定扩散模型可以在一般语言任务上与基于 AR 的 LLM 性能相匹配，同时在规划密集型场景中提供明显的优势，并通过任意顺序生成和动态质量速度优化提供前所未有的推理灵活性。我们发布了 Dream-Base 和 Dream-Instruct，以促进这一有希望的方向的进一步研究。

- 我们发布了 Dream 7B，它在各种基准测试中始终明显优于现有的扩散语言模型。
- Dream 7B 在标准基准测试（通用语言理解、数学推理和代码生成）上实现了与 Qwen 2.5 的竞争性能，同时展示了从扩散建模范式自然产生的卓越规划能力和新颖的推理灵活性功能。

2 相关工作

2.1 文本的扩散模型

扩散模型最初是为图像生成等连续领域开发的（Song 等人，2021；Ho 等人，2020）。在取得巨大成功的基础上，许多作品通过在连续嵌入空间中对文本进行建模、构建连续扩散语言模型，将扩散技术扩展到文本生成任务（Li et al., 2022；Gong et al., 2023b,a；Dieleman et al., 2022）。在这些方法中，Lin 等人。（2023）证明了在小型变压器的预训练和微调框架内应用扩散模型的可行性。

为了适应文本的离散性质，Austin 等人。（2021a）；霍格布姆等人。（2021）；坎贝尔等人。（2022）引入了离散扩散，它在离散词汇空间上运行，建立了

foundation for discrete diffusion methods. In this formulation, diffusion forward steps progressively with either [MASK] tokens (leading to an absorbing state) or random tokens (resulting in a uniform state), and the reverse process reconstructs the original text from these noised sequences. Building on this foundation, several studies have improved the training objectives for discrete diffusion (Zheng et al., 2024b; Lou et al., 2024; Shi et al., 2024; Ou et al., 2025; Zhao et al., 2024; Sahoo et al., 2024). Besides, He et al. (2023); Ye et al. (2025c) have shown benefits by initializing discrete diffusion models with pre-trained masked language models (MLMs, e.g., BERT; Devlin et al. 2019). Recent studies have highlighted diffusion language models’ advantages for complex reasoning and planning tasks (Ye et al., 2024, 2025a,b). Moreover, combining diffusion and autoregressive methods through block generation has emerged as a promising approach, demonstrating a flexible trade-off between parallelism and sequential coherence (Ye et al., 2024; Arriola et al., 2025).

2.2 Diffusion Large Language Models

Gulrajani and Hashimoto (2023a) analyzed the scaling laws for continuous diffusion language models, highlighting that continuous diffusion models require substantially longer training compared to autoregressive (AR) counterparts for optimal computational efficiency. In the discrete domain, Lou et al. (2024) showed that masked diffusion models (MDMs) could achieve perplexities comparable to or exceeding GPT-2-level AR models. Encouraged by these results, recent efforts have scaled diffusion language models to billions of parameters. Nie et al. (2025a) demonstrated the applicability of 1.1B-scale MDMs in language tasks such as question answering. Instead of training from scratch, Gong et al. (2025) proposed adapting existing AR language models (GPT-2 and LLaMA, from 127M to 7B parameters) into diffusion-based counterparts termed DiffuGPT and DiffuLLaMA. Concurrently, Nie et al. (2025b) presented LLaDA, an 8B-parameter diffusion language model trained from scratch, which achieved performance competitive with LLaMA3-8B (Grattafiori et al., 2024). Furthermore, Mercury Coder (Inception Labs, 2025) has recently shown commercial applicability and efficiency in code generation tasks. Overall, our work builds upon the masked diffusion paradigm and introduces new training techniques to push the performance further, highlighting the potential of diffusion-based models as an alternative to autoregressive LLMs.

3 Preliminary

3.1 Auto-regressive Modeling

Consider a sequence $\mathbf{x} := (\mathbf{x}^1, \dots, \mathbf{x}^N)$ of length N sampled from a data distribution $q(\mathbf{x})$. A fundamental approach in sequence modeling decomposes the joint probability of tokens into a product of conditional probabilities (Jelinek, 1980; Bengio et al., 2000):

$$p_{\theta}(\mathbf{x}) = p_{\theta}(\mathbf{x}^1) \underbrace{\prod_{n=2}^N p_{\theta}(\mathbf{x}^n | \mathbf{x}^{1:n-1})}_{\text{progressive left-context prediction}}, \quad (1)$$

where θ represents the model parameters and $\mathbf{x}^{1:n-1} := \mathbf{x}^1, \dots, \mathbf{x}^{n-1}$ denotes the preceding tokens.

3.2 Discrete Diffusion Modeling

Discrete diffusion models (Sohl-Dickstein et al., 2015; Hoogetboom et al., 2021; Austin et al., 2021a) are a class of latent variable models characterized by a forward noising process and a learned reverse denoising process. Let $\mathbf{x}_t$ denote the noised sequence at timestep t and the [MASK] noise is employed. The forward process $q(\mathbf{x}_{1:T} | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})$ progressively corrupts the original data $\mathbf{x}_0 := \mathbf{x}$ into a sequence of increasingly noisy tokens (replaced with [MASK] tokens) $\mathbf{x}_{1:T} := \mathbf{x}_1, \dots, \mathbf{x}_T$. The forward process continues until the sequence is fully denoised at $t = T$. For $t \in (0, T)$, the sequence $\mathbf{x}_t$ is partially noised using a noise schedule α_t (where for each token the probability of remaining unmasked is α_t). The backward process learns to gradually denoise the masked sequence back to the original data distribution by iteratively predicting masked tokens as t moves from T to 0. This denoising process can

离散扩散方法的基础。在此公式中，扩散向前逐步使用 [MASK] 标记（导致吸收状态）或随机标记（导致统一状态），而反向过程则从这些噪声序列中重建原始文本。在此基础上，一些研究改进了离散扩散的训练目标（Zheng et al., 2024b; Lou et al., 2024; Shi et al., 2024; Ou et al., 2025; Zhao et al., 2024; Sahoo et al., 2024）。此外，何等人。（2023）；叶等人。（2025c）已经展示了通过使用预训练的掩码语言模型（MLM，例如 BERT; Devlin 等人，2019）初始化离散扩散模型的好处。最近的研究强调了扩散语言模型在复杂推理和规划任务中的优势（Ye et al., 2024, 2025a,b）。此外，通过块生成将扩散和自回归方法相结合已成为一种有前途的方法，证明了并行性和顺序一致性之间的灵活权衡（Ye 等人，2024; Arriola 等人，2025）。

2.2 扩散大语言模型

Gulrajani 和 Hashimoto (2023a) 分析了连续扩散语言模型的缩放定律，强调与自回归 (AR) 模型相比，连续扩散模型需要更长的训练时间才能获得最佳计算效率。在离散域中，Lou 等人。（2024）表明，掩蔽扩散模型 (MDM) 可以实现与 GPT-2 级 AR 模型相当或超过的困惑度。受这些结果的鼓舞，最近的努力已将扩散语言模型扩展到数十亿个参数。聂等人。（2025a）证明了 1.1B 规模的 MDM 在问答等语言任务中的适用性。龚等人没有从头开始训练。（2025）提议将现有的 AR 语言模型（GPT-2 和 LLaMA，从 127M 到 7B 参数）调整为基于扩散的对应模型，称为 DiffuGPT 和 DiffuLLaMA。同时，聂等人。（2025b）提出了 LLaDA，这是一种从头开始训练的 8B 参数扩散语言模型，其性能可与 LLaMA3-8B 相媲美 (Grattafiori 等人，2024)。此外，Mercury Coder (Inception Labs, 2025) 最近在代码生成任务中展示了商业适用性和效率。总的来说，我们的工作建立在掩模扩散范式的基础上，并引入了新的训练技术来进一步提高性能，突出了基于扩散的模型作为自回归法学硕士替代方案的潜力。

3 初步

3.1 自回归建模

考虑从数据分布 $q(\mathbf{x})$ 采样的长度为 N 的序列 $\mathbf{x} := (\mathbf{x}^1, \dots, \mathbf{x}^N)$ 。序列建模的基本方法将标记的联合概率分解为条件概率的乘积 (Jelinek, 1980; Bengio 等人，2000)：

$$p_{\theta}(\mathbf{x}) = p_{\theta}(\mathbf{x}^1) \underbrace{\prod_{n=2}^N p_{\theta}(\mathbf{x}^n | \mathbf{x}^{1:n-1})}_{\text{progressive left-context prediction}}, \quad (1)$$

其中 θ 表示模型参数，而 $\mathbf{x}^{1:n-1} := \mathbf{x}^1, \dots, \mathbf{x}^{n-1}$ 表示前面的标记。

3.2 离散扩散建模

离散扩散模型 (Sohl-Dickstein et al., 2015; Hoogeboom et al., 2021; Austin et al., 2021a) 是一类潜变量模型，其特征是前向噪声过程和学习反向去噪过程。令 $\mathbf{x}_t$ 表示时间步 t 处的噪声序列，并采用 [MASK] 噪声。前向过程 $q(\mathbf{x}_{1:T} | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})$ 逐渐将原始数据 $\mathbf{x}_0 := \mathbf{x}$ 破坏为一系列噪声越来越大的标记（用 [MASK] 标记替换） $\mathbf{x}_{1:T} := \mathbf{x}_1, \dots, \mathbf{x}_T$ 。继续前向过程，直到序列在 $t = T$ 处完全去噪。对于 $t \in (0, T)$ ，序列 $\mathbf{x}_t$ 使用噪声调度 α_t （进行部分噪声处理，其中对于每个标记，保持未屏蔽的概率为 α_t ）。当 t 从 T 移动到 0 时，后向过程通过迭代预测屏蔽标记，学习将屏蔽序列逐渐去噪回到原始数据分布。这个去噪过程可以

be modeled as:

$$p_{\theta}(\mathbf{x}) = \sum_{\mathbf{x}_{1:T} \sim q} p(\mathbf{x}_T) \underbrace{\prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}_{\text{progressive full-context prediction}}. \quad (2)$$

The model parameters θ can be optimized by minimizing the negative log-likelihood of the clean data $\mathbf{x}_0$.

The discrete time formulation $t \in (0, T)$ restricts $\mathbf{x}_t$ to predetermined noise levels, which can introduce bias. To mitigate this limitation and enable flexible sampling across arbitrary noise levels, we adopt continuous-time parameterization where t ranges continuously over $[0, 1]$ (Kingma et al., 2021; Campbell et al., 2022; Shi et al., 2024; Ou et al., 2025). This continuous formulation allows the forward process to be expressed as $q(\mathbf{x}_t|\mathbf{x}_s)$ for any $0 \leq s < t \leq 1$. We employ discrete diffusion with absorbing states and simplify the objective as weighted cross-entropy losses:

$$L(\theta) = -\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}), t \sim \mathcal{U}(0,1), \mathbf{x}_t \sim q(\mathbf{x}_t|\mathbf{x}_0)} w(t) \sum_{n=1}^N \mathbf{1}_{[\mathbf{x}_t^n = \text{MASK}]} \log p_{\theta}(\mathbf{x}_0^n|\mathbf{x}_t), \quad (3)$$

where the indicator function $\mathbf{1}_{[\mathbf{x}_t^n = \text{MASK}]}$ ensures that the loss is computed only on masked token positions, and $w(t) \in (0, 1]$ is a time-dependent reweighting term determined by a transformation of α_t . For instance, when the forward process adopts the noise schedule $\alpha_t = 1 - t$, the corresponding weight becomes $w(t) = \frac{1}{t}$ (Austin et al., 2021a; Zheng et al., 2024b), which assigns greater importance to denoising steps closer to the clean data (i.e., as $t \rightarrow 0$). As demonstrated in Shi et al. (2024); Gong et al. (2025), this weighted cross-entropy objective Equation. (3) represents an effective reformulation of the ELBO, which enables optimization of the mask predictor $p_{\theta}(\cdot|\mathbf{x}_t)$ by providing a tractable variational upper bound on the negative log-likelihood.

4 Approach

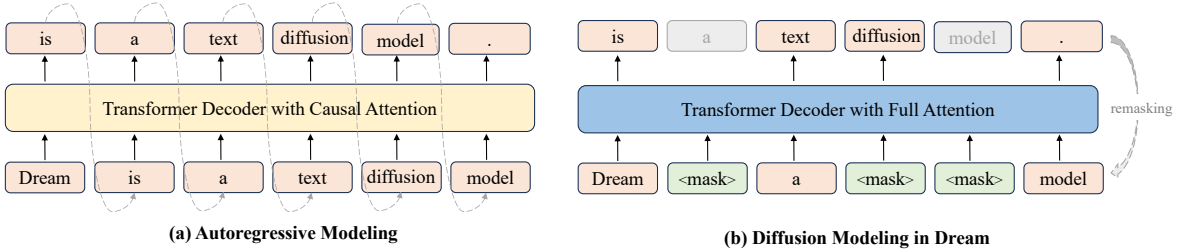


Figure 2: Comparison of autoregressive modeling and diffusion modeling in Dream. Dream predicts all the masked tokens in a shifted manner, allowing for maximum architectural alignment and weight initialization with AR models.

4.1 AR-based LLM Initialization

Building on DiffuLLaMA (Gong et al., 2025), we find that leveraging pre-trained weights from existing autoregressive (AR) models provides a non-trivial initialization strategy for diffusion language models. In Dream, we adopt the "Shift Operation" strategy in Gong et al. (2025) to maintain the inherent shift operation characteristic of AR models during the transition to diffusion-based training. Specifically, autoregressive models are trained so that the hidden representation at position i encodes the information necessary to predict the token at position $i + 1$. Rather than disrupting this learned positional relationship, our approach preserves this shift mechanism throughout the diffusion training process. The model continues to utilize hidden state h_i to generate predictions for position $i + 1$, which contrasts with conventional diffusion models that attempt to predict masked tokens at their original positions. This design choice ensures that the diffusion learning process remains aligned with the pretrained representations, preventing the transition from autoregressive to diffusion modeling from disrupting the foundational sequence understanding embedded in the pre-trained weights.

建模为：

$$p_{\theta}(\mathbf{x}) = \sum_{\mathbf{x}_{1:T} \sim q} p(\mathbf{x}_T) \underbrace{\prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)}_{\text{progressive full-context prediction}}. \quad (2)$$

可以通过最小化干净数据 $\mathbf{x}_0$ 的负对数似然来优化模型参数 θ 。

离散时间公式 $t \in (0, T)$ 将 $\mathbf{x}_t$ 限制为预定的噪声水平，这可能会引入偏差。为了减轻这一限制并实现跨任意噪声水平的灵活采样，我们采用连续时间参数化，其中 t 的范围连续超过 $[0, 1]$ （Kingma 等人，2021；Campbell 等人，2022；Shi 等人，2024；Ou 等人，2025）。对于任何 $0 \leq s < t \leq 1$ ，这种连续公式允许前向过程表示为 $q(\mathbf{x}_t | \mathbf{x}_s)$ 。我们采用具有吸收态的离散扩散，并将目标简化为加权交叉熵损失：

$$L(\theta) = -\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}), t \sim \mathcal{U}(0,1), \mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)} w(t) \sum_{n=1}^N \mathbf{1}_{[\mathbf{x}_t^n = \text{MASK}]} \log p_{\theta}(\mathbf{x}_0^n | \mathbf{x}_t), \quad (3)$$

其中指示函数 $\mathbf{1}_{[\mathbf{x}_t^n = \text{MASK}]}$ 确保仅在屏蔽标记位置上计算损失，而 $w(t) \in (0, 1]$ 是由 α_t 的变换确定的时间相关的重新加权项。例如，当前向过程采用噪声调度 $\alpha_t = 1 - t$ 时，相应的权重变为 $w(t) = \frac{1}{t}$ （Austin et al., 2021a；Cheng 等人，2024b），它更加重视更接近干净数据的去噪步骤（即，如 $t \rightarrow 0$ ）。如 Shi 等人（2024）；Gong 等人（2025）所示，这个加权交叉熵目标方程。（3）代表了 ELBO 的有效重新表述，可以优化掩模预测器 $p_{\theta}(\cdot | \mathbf{x}_t)$ 通过提供负对数似然的易处理的变分上限。

4 方法

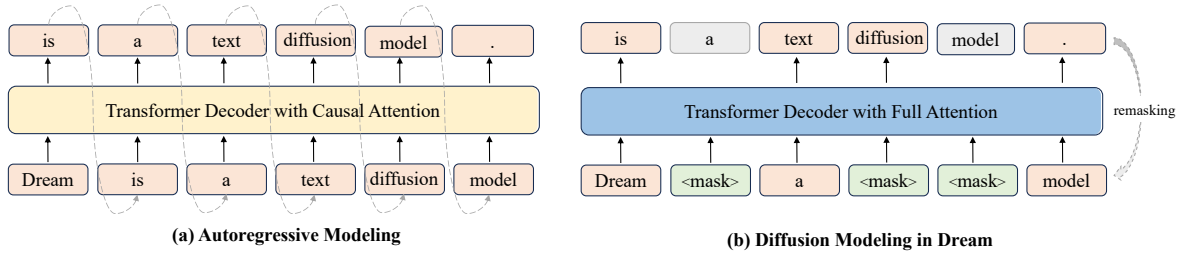


图 2：Dream 中自回归建模和扩散建模的比较。Dream 以转移的方式预测所有屏蔽标记，从而允许与 AR 模型进行最大程度的架构对齐和权重初始化。

4.1 基于AR的LLM初始化

在 DiffuLLaMA（Gong 等人，2025）的基础上，我们发现利用现有自回归（AR）模型的预训练权重为扩散语言模型提供了一种重要的初始化策略。在 Dream 中，我们采用了 Gong 等人的“转移操作”策略。（2025）在向基于扩散的训练过渡期间保持 AR 模型固有的移位操作特征。具体来说，训练自回归模型，以便位置 i 处的隐藏表示对预测位置 $i + 1$ 处的标记所需的信息进行编码。我们的方法不是破坏这种学习的位置关系，而是在整个扩散训练过程中保留了这种移位机制。该模型继续利用隐藏状态 h_i 来生成位置 $i + 1$ 的预测，这与尝试在其原始位置预测屏蔽标记的传统扩散模型形成对比。这种设计选择确保扩散学习过程与预训练表示保持一致，防止从自回归到扩散建模的过渡破坏预训练权重中嵌入的基本序列理解。

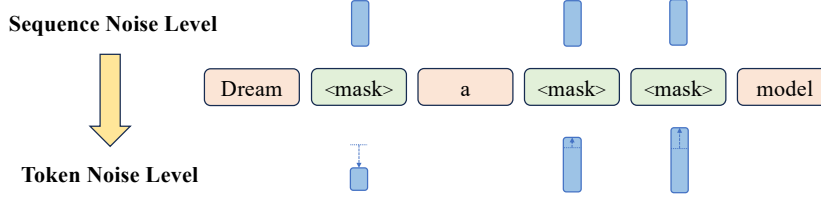


Figure 3: Illustration of the context-adaptive token-level noise rescheduling mechanism. Dream re-decides the noise level for each mask token by measuring its context informationness.

The AR initialization strategy offers computational advantages by building upon established sequence modeling capabilities rather than learning these representations from scratch, as evidenced by our experimental results in Section 5.4. Furthermore, this approach maximizes the utility of existing pre-trained AR language models. Once superior AR LLMs become available, our framework enables rapid iteration toward improved diffusion models without requiring costly from-scratch training, effectively leveraging the collective efforts of the research community in autoregressive model development.

4.2 Context-Adaptive Token-Level Noise Rescheduling

Standard diffusion training for text applies uniform sentence-level timesteps, which ignores the varying contextual dependencies of individual tokens. In conventional discrete diffusion training, a single timestep t is sampled to determine the overall noise level for the entire sequence, after which the model performs denoising on all the masked positions uniformly. For instance, when adopting the noise schedule $\alpha_t = 1 - t$, the weighting term $w(t)$ in Equation (3) becomes $\frac{1}{t}$. However, since learning occurs at the token level, the actual noise experienced by each token may deviate from the global timestep t . An example is shown in Figure 3, a shared sentence-level t is assigned to the three mask tokens in the standard discrete diffusion modeling, while the first token should inherently be associated with a lower noise level than subsequent tokens, as it benefits from richer contextual conditioning. This discrepancy leads to suboptimal learning, particularly for tokens with differing degrees of contextual information.

To address this limitation, we move from sequence-level noise scheduling to token-level noise scheduling. We introduce CART (**C**ontext-**A**daptive noise **R**escheduling at **T**oken-level), which measures the contextual informativeness for each token individually and dynamically reassigns appropriate noise levels based on the current context. This approach provides fine-grained, token-specific guidance that better reflects the varying difficulty and contextual dependencies across different positions in the sequence. Specifically, the objective with CART is represented as follows:

$$L(\theta) = -\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}), t \sim \mathcal{U}(0,1), \mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)} \sum_{n=1}^N \mathbf{1}_{[\mathbf{x}_t^n = \text{MASK}]} w(t, \mathbf{x}_t, n) \log p_{\theta}(\mathbf{x}_0^n | \mathbf{x}_t), \quad (4)$$

where $w(t, \mathbf{x}_t, n)$ is a generalized term of $w(t)$ that can be flexibly designed. In practice, we employ a mixture of geometric distributions to quantify the information contribution of each clean token relative to the noised tokens, which is defined as:

$$w(t, \mathbf{x}_t, n) = \frac{1}{2} \sum_{i=1}^N \mathbf{1}_{[\mathbf{x}_i \neq \text{MASK}]} \text{Geo}(p, |n - i| - 1), \quad (5)$$

where $p \in (0, 1]$ controls the sharpness of the geometric distribution. A smaller p causes clean tokens to tend to contribute uniformly to all masked tokens in the sequence, while a larger p forces clean tokens to have a greater influence on the nearby masked tokens.

4.3 Training

Model Architecture Dream models are built upon the Transformer architecture (Vaswani et al., 2017), maintaining full compatibility with existing autoregressive language models while enabling diffusion-based training. Specifically, Dream-7B adopts the same model configuration as Qwen2.5-7B (Qwen et al., 2025).

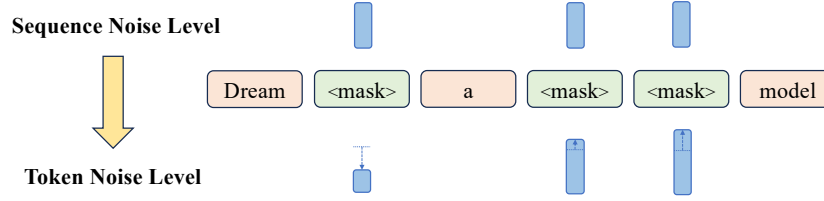


图 3: 上下文自适应令牌级噪声重新调度机制的图示。Dream 通过测量其上下文信息来重新确定每个掩码标记的噪声级别。

AR 初始化策略通过构建已建立的序列建模功能而不是从头开始学习这些表示来提供计算优势，正如我们在第 5.4 节中的实验结果所证明的那样。此外，这种方法最大化了现有预训练 AR 语言模型的效用。一旦高级 AR LLM 可用，我们的框架就可以快速迭代改进扩散模型，而无需昂贵的从头开始培训，从而有效地利用研究社区在自回归模型开发方面的集体努力。

4.2 上下文自适应令牌级噪声重新调度

文本的标准扩散训练应用统一的句子级时间步长，它忽略了各个标记的不同上下文依赖性。在传统的离散扩散训练中，对单个时间步长 t 进行采样以确定整个序列的总体噪声水平，之后模型对所有屏蔽位置均匀地执行去噪。例如，当采用噪声调度 $\alpha_t = 1 - t$ 时，等式(3)中的加权项 $w(t)$ 变为 $\frac{1}{t}$ 。然而，由于学习发生在令牌级别，因此每个令牌经历的实际噪声可能会偏离全局时间步长 t 。图 3 中显示了一个示例，共享句子级 t 被分配给标准离散扩散模型中的三个掩码标记，而第一个标记本质上应该与比后续标记更低的噪声水平相关联，因为它受益于更丰富的上下文调节。这种差异会导致学习效果不佳，特别是对于具有不同程度上下文信息的标记。

为了解决这个限制，我们从序列级噪声调度转向令牌级噪声调度。我们引入了 CART（令牌级别的上下文自适应噪声重新调度），它单独测量每个令牌的上下文信息，并根据当前上下文动态地重新分配适当的噪声级别。这种方法提供了细粒度的、特定于令牌的指导，可以更好地反映序列中不同位置的不同难度和上下文依赖性。具体来说，CART 的目标表示如下：

$$L(\theta) = -\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}), t \sim \mathcal{U}(0,1), \mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)} \sum_{n=1}^N \mathbf{1}_{[\mathbf{x}_t^n = \text{MASK}]} w(t, \mathbf{x}_t, n) \log p_{\theta}(\mathbf{x}_0^n | \mathbf{x}_t), \quad (4)$$

其中 $w(t, \mathbf{x}_t, n)$ 是 $w(t)$ 的泛化项，可以灵活设计。在实践中，我们采用几何分布的混合来量化每个干净标记相对于噪声标记的信息贡献，其定义为：

$$w(t, \mathbf{x}_t, n) = \frac{1}{2} \sum_{i=1}^N \mathbf{1}_{[\mathbf{x}_i^n \neq \text{MASK}]} \text{Geo}(p, |n - i| - 1), \quad (5)$$

其中 $p \in (0, 1]$ 控制几何分布的锐度。较小的 p 会导致干净的令牌倾向于对序列中的所有屏蔽令牌做出均匀的贡献，而较大的 p 会迫使干净的令牌对附近的屏蔽令牌产生更大的影响。

4.3 培训

模型架构 Dream 模型建立在 Transformer 架构之上 (Vaswani et al., 2017)，保持与现有自回归语言模型的完全兼容性，同时实现基于扩散的训练。具体来说，Dream-7B 采用与 Qwen2.5-7B 相同的模型配置 (Qwen et al., 2025)。

Pretraining The training objective is to model the data distribution $p_\theta(\mathbf{x}_0)$. We estimate Equation (3) via the Monte Carlo method for stochastic gradient descent training. Our training corpus comprises 580 billion tokens spanning text, mathematics, and code domains. The data is sourced from three primary repositories: Dolma v1.7 (Soldaini et al., 2024) for general domain text, OpenCoder (Huang et al., 2025) for programming and code datasets, and DCLM-Baseline (Li et al., 2024) for curated high-quality text and mathematical content. We exclusively utilize open-source datasets to demonstrate the effectiveness of our paradigm and ensure reproducibility.

Supervised Fine-Tuning The SFT training objective is to model the conditional distribution $p_\theta(r_0|p_0)$, where p_0 is the prompt and r_0 denotes the response. During fine-tuning, noise is applied only to the response portion of the sequence, keeping the prompt intact to preserve instruction following capabilities. To align Dream with user instructions, we curate a dataset containing 1.8M instruction-response pairs from Tulu 3 (Lambert et al., 2025) and SmoLLM 2 (Allal et al., 2025).

5 Experiments

5.1 Setup

Evaluation Tasks We evaluated Dream on a comprehensive suite of benchmarks designed to assess its capabilities across multiple domains. For general language understanding, we employed tasks such as MMLU (Hendrycks et al., 2020), BBH (Suzgun et al., 2022), ARC-E (Clark et al., 2018), ARC-C (Clark et al., 2018), HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2021), PIQA (Bisk et al., 2020), and RACE (Lai et al., 2017) to test knowledge, reasoning, and commonsense abilities. For mathematical and scientific reasoning, we used GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2020) for numerical computation and problem-solving, and GPQA (Rein et al., 2023) for scientific understanding. Code generation abilities were evaluated through HumanEval (Chen et al., 2021) and MBPP (Austin et al., 2021b) to measure programming capabilities. The IFEval benchmark specifically measured the model’s capacity to adhere to given instructions for instruction following assessment. For planning tasks, we evaluate performance on the Countdown and Sudoku tasks from Ye et al. (2025a), and Trip planning task from Zheng et al. (2024a), which assess planning capabilities with specific objectives.

5.2 Results of Dream-Base

Superior planning performance of diffusion language models over autoregressive models

As shown in Table 1, Dream 7B maintains competitive general language capabilities while excelling in complex reasoning scenarios compared to Qwen2.5 7B, demonstrating the superior reasoning and planning abilities of diffusion models. For instance, on general tasks, Dream 7B achieves comparable performance with only modest gaps on MMLU and HellaSwag, while outperforming on reasoning tasks (ARC-E, ARC-C). The most striking advantages emerge in planning tasks, where Dream 7B substantially outperforms Qwen2.5 on Countdown (16.0 vs. 6.2), Sudoku (81.0 vs. 21.0), and Trip planning (17.8 vs. 3.6). Furthermore, we observe that other diffusion models (LLaDA) also consistently outperform Qwen2.5 on these tasks, providing additional evidence for the inherent advantages of diffusion modeling in planning capabilities.

Effectiveness of Dream training approach

Compared to the previous state-of-the-art diffusion model (LLaDA 8B, which is trained from scratch), Dream 7B achieves remarkable performance improvements across all evaluation domains while using only 1/4 of the training data (0.6T vs 2.3T tokens). Dream 7B demonstrates significant gains on general tasks, with even more pronounced advantages in reasoning tasks. Notably, Dream 7B achieves substantial improvements on reasoning tasks (e.g., ARC-C improves from 47.5 to 59.8, GSM8K improves from 70.9 to 77.2). Additionally, Dream 7B excels in planning tasks with substantial margins over LLaDA: Countdown improves from 13.2 to 16.0, and Sudoku from 46.0 to 81.0. These results validate the effectiveness of our autoregressive model initialization strategy and context-adaptive noise rescheduling mechanism. Notably, since Dream 7B is initialized from Qwen2.5 weights, we expect that with stronger autoregressive foundation models, our approach will enable the development of even more capable diffusion language models.

预训练 训练目标是对数据分布 $p_{\theta}(\mathbf{x}_0)$ 进行建模。我们通过随机梯度下降训练的蒙特卡罗方法估计方程 (3)。我们的训练语料库包含 5800 亿个令牌，涵盖文本、数学和代码领域。数据来源于三个主要存储库：用于通用领域文本的 Dolma v1.7 (Soldaini 等人, 2024)、用于编程和代码数据集的 OpenCoder (Huang 等人, 2025) 以及用于精选高质量文本和数学内容的 DCLM-Baseline (Li 等人, 2024)。我们专门利用开源数据集来证明我们范例的有效性并确保可重复性。

有监督微调 SFT 训练目标是对条件分布 $p_{\theta}(r_0|p_0)$ 进行建模，其中 p_0 是提示， r_0 表示响应。在微调期间，噪声仅应用于序列的响应部分，保持提示完整以保留指令跟随能力。为了使 Dream 与用户指令保持一致，我们整理了一个数据集，其中包含来自 Tulu 3 (Lambert 等人, 2025) 和 SmolLM 2 (Allal 等人, 2025) 的 180 万个指令响应对。

5 实验

5.1 设置

评估任务 我们根据一套全面的基准对 Dream 进行了评估，这些基准旨在评估其跨多个领域的功能。对于一般语言理解，我们采用了 MMLU (Hendrycks et al., 2020)、BBH (Suzgun et al., 2022)、ARC-E (Clark et al., 2018)、ARC-C (Clark et al., 2018)、HellaSwag (Zellers et al., 2019)、WinoGrande (Sakaguchi et al., 2019) 等任务。2021)、PIQA (Bisk 等人, 2020) 和 RACE (Lai 等人, 2017) 来测试知识、推理和常识能力。对于数学和科学推理，我们使用 GSM8K (Cobbe et al., 2021) 和 MATH (Hendrycks et al., 2020) 进行数值计算和问题解决，并使用 GPQA (Rein et al., 2023) 进行科学理解。通过 HumanEval (Chen et al., 2021) 和 MBPP (Austin et al., 2021b) 评估代码生成能力，以衡量编程能力。IFEval 基准专门衡量模型遵循给定指令进行评估的能力。对于规划任务，我们评估了 Ye 等人的倒计时和数独任务的性能。(2025a)，以及郑等人的旅行规划任务。(2024a)，评估具有特定目标的规划能力。

5.2 梦想基地成果

扩散语言模型优于自回归模型的规划性能如表1所示，与Qwen2.5 7B相比，Dream 7B在保持有竞争力的通用语言能力的同时，在复杂推理场景中表现出色，展现了扩散模型优越的推理和规划能力。例如，在一般任务上，Dream 7B 在 MMLU 和 HellaSwag 上实现了可比的性能，仅略有差距，而在推理任务（ARC-E、ARC-C）上表现优于。最显著的优势出现在规划任务中，其中 Dream 7B 在倒计时（16.0 比 6.2）、数独（81.0 比 21.0）和旅行规划（17.8 比 3.6）方面明显优于 Qwen2.5。此外，我们观察到其他扩散模型 (LLaDA) 在这些任务上也始终优于 Qwen2.5，这为扩散建模在规划能力方面的固有优势提供了额外的证据。

Dream 训练方法的有效性与之前最先进的扩散模型 (LLaDA 8B，从头开始训练) 相比，Dream 7B 在仅使用 1/4 的训练数据 (0.6T 与 2.3T 令牌) 的情况下在所有评估领域实现了显著的性能改进。Dream 7B 在一般任务上表现出了显著的进步，在推理任务上的优势更为明显。值得注意的是，Dream 7B 在推理任务上取得了实质性的改进（例如，ARC-C 从 47.5 提高到 59.8，GSM8K 从 70.9 提高到 77.2）。此外，Dream 7B 在规划任务方面表现出色，与 LLaDA 相比有很大优势：倒计时从 13.2 提高到 16.0，数独从 46.0 提高到 81.0。这些结果验证了我们的自回归模型初始化策略和上下文自适应噪声重新调度机制的有效性。值得注意的是，由于 Dream 7B 是从 Qwen2.5 权重初始化的，我们预计通过更强大的自回归基础模型，我们的方法将能够开发出功能更强大的扩散语言模型。

Table 1: Comparison of base models on standard evaluation benchmarks. * indicates Dream 7B, LLaDA 8B, Qwen2.5 7B and LLaMA3 8B are evaluated under the same protocol. The best results are bolded and the second best are underlined. Our results are highlighted in blue.

Model	Dream 7B*	LLaDA 8B*	Qwen2.5 7B*	LLaMA3 8B*	Mistral 7B	DeepSeek 7B
Type	Diffusion	Diffusion	AR	AR	AR	AR
Training Tokens	0.6T	2.3T	18T	15T	-	2T
General Tasks						
MMLU	<u>69.5 (5)</u>	65.9 (5)	71.9 (5)	63.5 (5)	60.1 (5)	48.2 (5)
BBH	<u>57.9 (3)</u>	47.4 (3)	63.9 (3)	62.7 (3)	-	39.5 (3)
ARC-E	83.9 (0)	71.8 (0)	77.4 (0)	<u>81.1 (0)</u>	80.0 (0)	67.9 (0)
ARC-C	59.8 (0)	47.5 (0)	51.5 (0)	<u>53.6 (0)</u>	55.5 (0)	48.1 (0)
Hellaswag	<u>73.3 (0)</u>	72.7 (0)	79.0 (0)	<u>78.9 (0)</u>	81.3 (0)	75.4 (0)
WinoGrande	<u>74.5 (5)</u>	73.5 (5)	<u>76.4 (5)</u>	76.9 (5)	75.3 (0)	70.5 (0)
PIQA	<u>75.8 (0)</u>	74.8 (0)	<u>79.8 (0)</u>	81.3 (0)	83.0 (0)	79.2 (0)
RACE	44.7 (0)	38.7 (0)	<u>41.9 (0)</u>	39.2 (0)	-	46.5 (5)
Mathematics & Science						
GSM8K	<u>77.2 (8)</u>	70.9 (8)	78.9 (8)	55.3 (8)	52.1 (8)	17.4 (8)
MATH	<u>39.6 (4)</u>	30.7 (4)	41.1 (4)	18.0 (4)	13.1 (4)	6.0 (4)
GPQA	36.6 (5)	30.4 (5)	<u>35.5 (5)</u>	30.6 (5)	-	-
Code						
HumanEval	57.9 (0)	32.9 (0)	<u>56.7 (0)</u>	35.4 (0)	30.5 (0)	26.2 (0)
MBPP	<u>56.2 (4)</u>	39.0 (4)	63.6 (4)	49.2 (4)	47.5 (3)	39.0 (3)
Planning Tasks						
Countdown	16.0 (8)	<u>13.2 (8)</u>	6.2 (8)	3.7 (8)	-	-
Sudoku	81.0 (8)	<u>46.0 (8)</u>	21.0 (8)	0.0 (8)	-	-
Trip planning	17.8 (2)	<u>16.4 (2)</u>	3.6 (2)	8.7 (2)	-	-

5.3 Results of Dream-Instruct

As an early exploration of diffusion LLM post-training, we perform lightweight supervised fine-tuning to align Dream with user instructions. Specifically, we curate 1.8M instruction-response pairs and fine-tune Dream for 3 epochs. The results of Dream-Instruct in Table 2 demonstrate DLLM’s potential to match AR-based LLMs in instruction-following tasks, establishing a foundation for future advanced DLLM post-training recipes.

Table 2: Performance comparison after supervised fine-tuning on standard evaluation benchmarks.

Model	Dream 7B	LLaDA 8B	Qwen2.5 7B	LLaMA3 8B
Type	Diffusion	Diffusion	AR	AR
Post-training	SFT	SFT	SFT+RL	SFT+RL
Alignment pairs	1.8M	4.5M	1M/0.15M	-
MMLU	67.0	65.5	76.6	68.4
MMLU-pro	43.3	37	56.3	41.9
GSM8K	81.0	78.6	91.6	78.3
Math	39.2	26.6	75.5	29.6
GPQA	33.0	31.8	36.4	31.9
HumanEval	55.5	47.6	84.8	59.8
MBPP	58.8	34.2	79.2	57.6
IFEval	62.5	59.9	74.7	49.7

表 1: 基本模型在标准评估基准上的比较。* 表示 Dream 7B、LLaDA 8B、Qwen2.5 7B 和 LLaMA3 8B 在相同协议下进行评估。最好的结果加粗，次好的结果加下划线。我们的结果以蓝色突出显示。

Model	Dream 7B*	LLaDA 8B*	Qwen2.5 7B*	LLAMA3 8B*	Mistral 7B	DeepSeek 7B
Type	Diffusion	Diffusion	AR	AR	AR	AR
Training Tokens	0.6T	2.3T	18T	15T	-	2T
General Tasks						
MMLU	69.5 (5)	65.9 (5)	71.9 (5)	63.5 (5)	60.1 (5)	48.2 (5)
BBH	57.9 (3)	47.4 (3)	63.9 (3)	62.7 (3)	-	39.5 (3)
ARC-E	83.9 (0)	71.8 (0)	77.4 (0)	81.1 (0)	80.0 (0)	67.9 (0)
ARC-C	59.8 (0)	47.5 (0)	51.5 (0)	53.6 (0)	55.5 (0)	48.1 (0)
Hellaswag	73.3 (0)	72.7 (0)	79.0 (0)	78.9 (0)	81.3 (0)	75.4 (0)
WinoGrande	74.5 (5)	73.5 (5)	76.4 (5)	76.9 (5)	75.3 (0)	70.5 (0)
PIQA	75.8 (0)	74.8 (0)	79.8 (0)	81.3 (0)	83.0 (0)	79.2 (0)
RACE	44.7 (0)	38.7 (0)	41.9 (0)	39.2 (0)	-	46.5 (5)
Mathematics & Science						
GSM8K	77.2 (8)	70.9 (8)	78.9 (8)	55.3 (8)	52.1 (8)	17.4 (8)
MATH	39.6 (4)	30.7 (4)	41.1 (4)	18.0 (4)	13.1 (4)	6.0 (4)
GPQA	36.6 (5)	30.4 (5)	35.5 (5)	30.6 (5)	-	-
Code						
HumanEval	57.9 (0)	32.9 (0)	56.7 (0)	35.4 (0)	30.5 (0)	26.2 (0)
MBPP	56.2 (4)	39.0 (4)	63.6 (4)	49.2 (4)	47.5 (3)	39.0 (3)
Planning Tasks						
Countdown	16.0 (8)	13.2 (8)	6.2 (8)	3.7 (8)	-	-
Sudoku	81.0 (8)	46.0 (8)	21.0 (8)	0.0 (8)	-	-
Trip planning	17.8 (2)	16.4 (2)	3.6 (2)	8.7 (2)	-	-

5.3 梦授的结果

作为扩散 LLM 后期训练的早期探索，我们进行轻量级监督微调，以使 Dream 与用户指令保持一致。具体来说，我们策划了 180 万个指令响应对，并对 Dream 进行了 3 个时期的微调。表 2 中 Dream-Instruct 的结果证明了 DLLM 在指令跟踪任务中与基于 AR 的 LLM 相匹配的潜力，为未来高级 DLLM 后训练方法奠定了基础。

表 2: 在标准评估基准上进行监督微调后的性能比较。

Model	Dream 7B	LLaDA 8B	Qwen2.5 7B	LLaMA3 8B
Type	Diffusion	Diffusion	AR	AR
Post-training	SFT	SFT	SFT+RL	SFT+RL
Alignment pairs	1.8M	4.5M	1M/0.15M	-
MMLU	67.0	65.5	76.6	68.4
MMLU-pro	43.3	37	56.3	41.9
GSM8K	81.0	78.6	91.6	78.3
Math	39.2	26.6	75.5	29.6
GPQA	33.0	31.8	36.4	31.9
HumanEval	55.5	47.6	84.8	59.8
MBPP	58.8	34.2	79.2	57.6
IFEval	62.5	59.9	74.7	49.7

5.4 Effect of AR Initialization

To validate the effectiveness of autoregressive (AR) initialization, we conducted preliminary experiments adapting Dream-1B from LLaMA3.2-1B. As demonstrated in Figure 4, the AR initialization strategy proves particularly effective during the early stages of training, providing significant advantages over training from scratch. This observation shows that the existing left-to-right knowledge in AR models provides a strong foundation that accelerates the diffusion model’s development of any-order generation capabilities. This acceleration significantly reduces both the number of tokens and computational resources required for pretraining, making our approach highly efficient compared to training diffusion language models from scratch.

Besides, our analysis reveals that the learning rate plays a critical role in preserving the beneficial properties inherited from AR initialization. An excessively high learning rate can rapidly degrade the left-to-right linguistic knowledge embedded in the initial weights, thereby diminishing the advantages of AR initialization. Conversely, an overly conservative learning rate may impede the model’s ability to effectively learn the diffusion process. Through systematic hyperparameter tuning, we carefully calibrated the learning rate alongside other training parameters to achieve optimal performance.

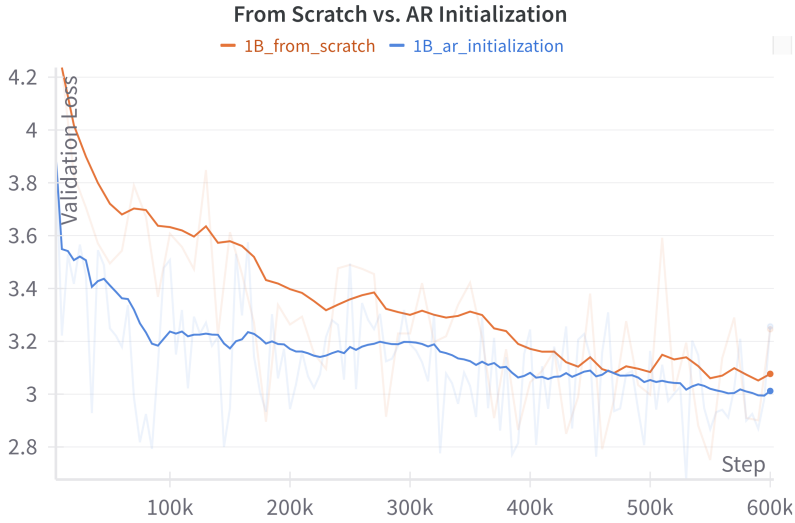


Figure 4: Loss comparison of the from-scratch and AR-initialization with LLaMA3.2 1B training on the Dream 1B models with 200B tokens. AR initialization also experiences a high loss at the beginning due to the transition from causal attention to full attention; however, this loss remains lower compared to training from scratch throughout the training.

5.5 Discussion

5.5.1 Analysis of Planning Ability

Ye et al. 2025a,b demonstrated that text diffusion models exhibit superior planning capabilities in small-scale, task-specific contexts. However, it remained unclear whether these advantages extend to general-purpose, scaled diffusion models. In this work, we scale up diffusion language models and systematically evaluate their constraint-based planning abilities using Dream 7B. We evaluated Dream on the Countdown and Sudoku tasks from Ye et al. (2025a). Our experimental comparison included Dream 7B alongside LLaDA 8B, Qwen2.5 7B, and LLaMA3 8B, with the DeepSeek V3-671B (0324) serving as a reference point. All models were evaluated in a few-shot setting without task-specific training.

The results shown in Figure 5 demonstrate that Dream consistently outperforms other similarly-sized baseline models across both tasks. Notably, Dream 7B significantly exceeds the performance of comparably-sized autoregressive models (LLaMA3 8B and Qwen2.5 7B) and, in certain cases (e.g., Countdown3), even surpasses the much larger DeepSeek-V3-671B (0324), despite the latter having

5.4 AR初始化的效果

为了验证自回归 (AR) 初始化的有效性，我们采用 LLaMA3.2-1B 中的 Dream-1B 进行了初步实验。如图 4 所示，事实证明，AR 初始化策略在训练的早期阶段特别有效，与从头开始训练相比具有显著优势。这一观察结果表明，AR 模型中现有的从左到右的知识为加速扩散模型任意阶生成能力的发展提供了坚实的基础。这种加速显著减少了预训练所需的标记数量和计算资源，使我们的方法比从头开始训练扩散语言模型更加高效。

此外，我们的分析表明，学习率在保留从 AR 初始化继承的有益属性方面起着至关重要的作用。过高的学习率会迅速降低初始权重中嵌入的从左到右的语言知识，从而削弱 AR 初始化的优势。相反，过于保守的学习率可能会阻碍模型有效学习扩散过程的能力。通过系统的超参数调整，我们仔细校准了学习率以及其他训练参数，以实现最佳性能。

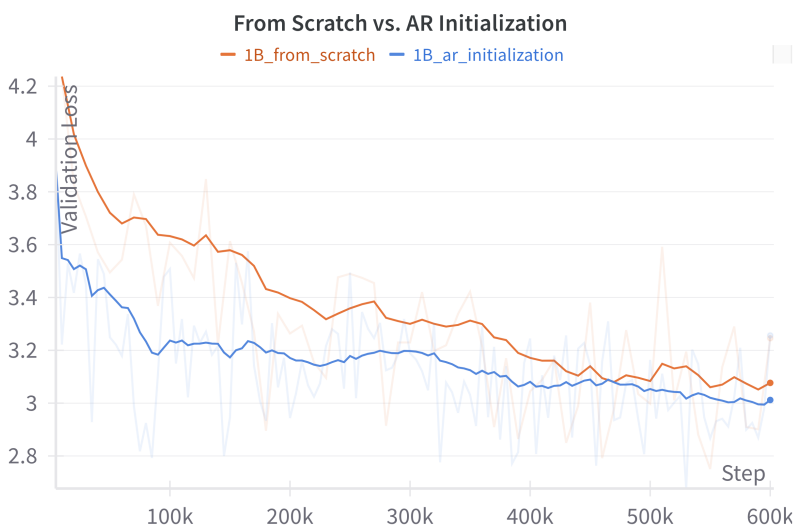


图 4：从头开始和 AR 初始化与 LLaMA3.2 1B 训练在具有 200B 代币的 Dream 1B 模型上的损失比较。由于从因果注意力到完全注意力的转变，AR 初始化在开始时也会经历很高的损失；然而，与整个训练期间从头开始训练相比，这种损失仍然较低。

5.5 讨论

5.5.1 规划能力分析

叶等人。2025a,b 证明文本传播模型在小规模、特定任务的环境中表现出卓越的规划能力。然而，目前尚不清楚这些优势是否可以扩展到通用的、按比例缩放的扩散模型。在这项工作中，我们扩展了扩散语言模型，并使用 Dream 7B 系统地评估了它们基于约束的规划能力。我们对 Ye 等人的倒计时和数独任务中的 Dream 进行了评估。（2025a）。我们的实验比较包括 Dream 7B 以及 LLaDA 8B、Qwen2.5 7B 和 LLaMA3 8B，并以 DeepSeek V3-671B (0324) 作为参考点。所有模型均在几次设置中进行评估，无需进行特定任务的训练。

图 5 中显示的结果表明，Dream 在这两项任务中始终优于其他类似大小的基线模型。值得注意的是，Dream 7B 显著超过了同等大小的自回归模型（LLaMA3 8B 和 Qwen2.5 7B）的性能，并且在某些情况下（例如 Countdown3），甚至超过了更大的 DeepSeek-V3-671B (0324)，尽管后者具有

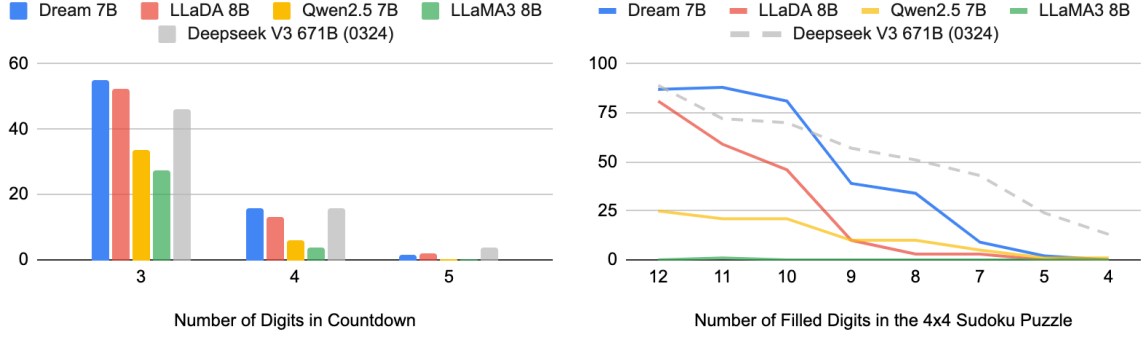


Figure 5: Results on Countdown and Sudoku when varying planning difficulty.

orders of magnitude more parameters. This substantial performance advantage suggests that diffusion language models are inherently more effective at solving problems involving multiple constraints or specific objective optimization. For detailed qualitative examples illustrating the performance differences between Qwen2.5 7B and Dream 7B across various planning scenarios, please refer to Figure 6.

Task	Question (In-context examples are omitted)	Answer from Qwen2.5 7B (AR)	Answer from Dream 7B (Diffusion)
Trip Planning	You plan to visit 3 European cities for 12 days in total. You only take direct flights to commute between cities. You would like to visit Istanbul for 5 days. You would like to visit Budapest for 6 days. You would like to visit Dubrovnik for 3 days.	Here is the trip plan for visiting the 3 European cities for 12 days: **Day 1-5:** Arriving in Istanbul and visit Istanbul for 5 days. **Day 5:** Fly from Istanbul to Budapest. **Day 5-11:** Visit Budapest for 6 days. **Day 11:** Fly from Budapest to Dubrovnik. **Day 11-12:** Visit Dubrovnik for 3 days.	Here is the trip plan for visiting the 3 European cities for 12 days: **Day 1-3:** Arriving in Dubrovnik and visit Dubrovnik for 3 days. **Day 3:** Fly from Dubrovnik to Istanbul. **Day 3-7:** Visit Istanbul for 5 days. **Day 7:** Fly from Istanbul to Budapest. **Day 7-12:** Visit Budapest for 6 days.
	Here are the cities that have direct flights: Istanbul and Budapest, from Dubrovnik to Istanbul.		
	Find a trip plan of visiting the cities for 12 days by taking direct flights to commute between them.		
Countdown	Given 5 numbers, use +-* / to operate over the first four numbers to achieve the fifth number. Input: 81,84,82,42,41 Output:	84-81=3,82-42=40,40/3=13.3333333	82-81=1,84-1=83,83-42=41
	Fill the position where the values are 0 in a 4x4 grid with digits 1-4 so that each column, each row, and each of the four 2x2 subgrids that compose the grid contains all of the digits from 1 to 4. Input: 4001 3124 1302 0400 Output:	4312 2134 1342 3241	4231 3124 1342 2413

Figure 6: Examples generated by Qwen2.5 7B and Dream 7B.

5.5.2 Quality-Speed Trade-Offs

Diffusion language models provide a unique advantage through their adjustable inference process, where the number of diffusion timesteps can be modified to balance computational speed and output quality. Reducing the number of timesteps speeds up inference but may reduce generation quality, while increasing timesteps improves output quality at the cost of additional computation.

To demonstrate this flexibility, we conduct a systematic quality-speed analysis on the Countdown task, comparing Dream 7B against Qwen2.5 7B across varying timestep configurations, as illustrated in Figure 7. The experimental results demonstrate that Dream’s performance exhibits a controllable trade-off between computational efficiency and generation quality through timestep adjustment, providing users with fine-grained control over the inference process. Notably, when diffusion steps are set between 5-20, Dream achieves superior performance in both speed and quality compared to Qwen2.5 7B.

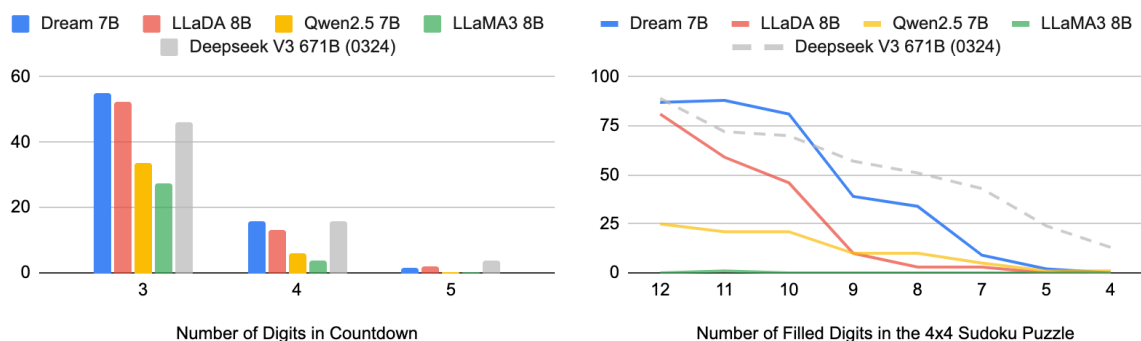


图 5：不同计划难度时倒计时和数独的结果。

更多数量级的参数。这种巨大的性能优势表明，扩散语言模型本质上在解决涉及多个约束或特定目标优化的问题时更有效。有关说明 Qwen2.5 7B 和 Dream 7B 在各种规划场景中的性能差异的详细定性示例，请参阅图 6。

Task	Question (In-context examples are omitted)	Answer from Qwen2.5 7B (AR)	Answer from Dream 7B (Diffusion)
Trip Planning	You plan to visit 3 European cities for 12 days in total. You only take direct flights to commute between cities. You would like to visit Istanbul for 5 days. You would like to visit Budapest for 6 days. You would like to visit Dubrovnik for 3 days.	Here is the trip plan for visiting the 3 European cities for 12 days: **Day 1-5:** Arriving in Istanbul and visit Istanbul for 5 days. **Day 5-11:** Fly from Istanbul to Budapest. Visit Budapest for 6 days. **Day 11-12:** Fly from Budapest to Dubrovnik. **Day 11-12:** Visit Dubrovnik for 3 days.	Here is the trip plan for visiting the 3 European cities for 12 days: **Day 1-3:** Arriving in Dubrovnik and visit Dubrovnik for 3 days. **Day 3-7:** Fly from Dubrovnik to Istanbul. Visit Istanbul for 5 days. **Day 7-12:** Fly from Istanbul to Budapest. Visit Budapest for 6 days.
	Here are the cities that have direct flights: Istanbul and Budapest, from Dubrovnik to Istanbul.		
	Find a trip plan of visiting the cities for 12 days by taking direct flights to commute between them.		
Countdown	Given 5 numbers, use +-* / to operate over the first four numbers to achieve the fifth number. Input: 81,84,82,42,41		
	Output:	84-81=3,82-42=40,40/3=13.3333333	82-81=1,84-1=83,83-42=41
Sudoku	Fill the position where the values are 0 in a 4x4 grid with digits 1-4 so that each column, each row, and each of the four 2x2 subgrids that compose the grid contains all of the digits from 1 to 4.		
	Input:		
	4001	4312	4231
	3124	2134	3124
	1302	1342	1342
	0400	3241	2413
	Output:		

图 6：Qwen2.5 7B 和 Dream 7B 生成的示例。

5.5.2 质量与速度的权衡

扩散语言模型通过其可调整的推理过程提供了独特的优势，其中可以修改扩散时间步数以平衡计算速度和输出质量。减少时间步数可以加快推理速度，但可能会降低生成质量，而增加时间步数可以提高输出质量，但会增加额外计算的成本。

为了证明这种灵活性，我们对 Countdown 任务进行了系统的质量-速度分析，在不同的时间步配置下将 Dream 7B 与 Qwen2.5 7B 进行了比较，如图 7 所示。实验结果表明，Dream 的性能通过时间步调整在计算效率和生成质量之间呈现出可控的权衡，为用户提供了对推理过程的细粒度控制。值得注意的是，当扩散步数设置在 5-20 之间时，与 Qwen2.5 7B 相比，Dream 在速度和质量上都取得了优异的性能。

This timestep-based approach introduces an additional dimension for inference-time scaling (Snell et al., 2024; Muennighoff et al., 2025; Geiping et al., 2025), which works alongside existing techniques such as chain-of-thought reasoning used in large language models like OpenAI o1 (Jaech et al., 2024) and DeepSeek R1 (Guo et al., 2025). This adjustable computation-quality trade-off represents a key advantage that sets diffusion models apart from traditional autoregressive models.

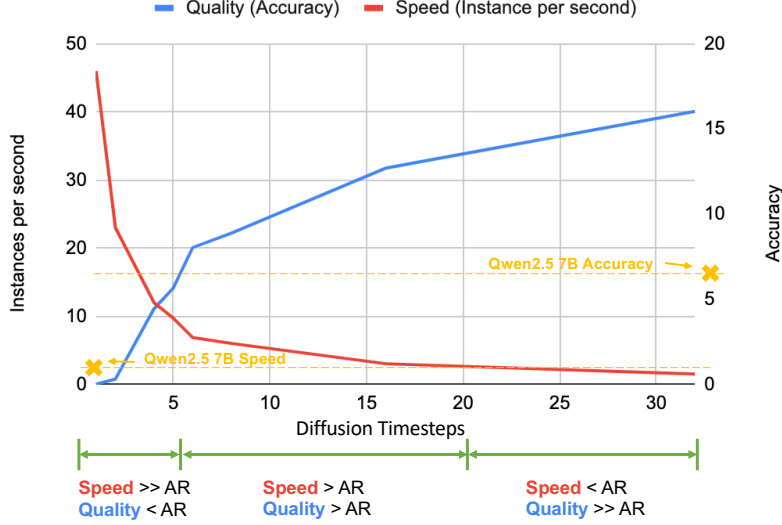


Figure 7: Quality-speed comparison on the Countdown task for Dream 7B and Qwen2.5 7B. By adjusting the diffusion timesteps, the performance of Dream can be flexibly tuned for either speed or quality.

5.5.3 Arbitrary Order Generation

Diffusion models exhibit remarkable inference flexibility, setting them apart from traditional autoregressive (AR) models that typically generate sequences in a fixed left-to-right order. This flexibility stems from their inherent capability to synthesize outputs in arbitrary orders, opening up a wider range of diverse user queries and interaction patterns.

Dream 7B supports completion and infilling tasks naturally, without requiring specialized training. Key examples of this flexibility include:

- **Completion:** Users can provide an initial piece of text, and the model can seamlessly continue or complete it.
- **Infilling:** Diffusion models can fill in missing segments within a given text. This can be unconstrained or guided by specific requirements, such as ensuring the generated infill leads to an exact ending sentence.
- **Configurable decoding order:** Beyond these task-specific flexibilities, users can also exert control over the decoding behavior itself. By adjusting decoding hyperparameters, the generation process can be tailored to mimic the more structured, left-to-right generation characteristic of AR models, or conversely, it can introduce varying degrees of randomness in the decoding order, moving towards a partially or even fully random-order synthesis.

This adaptability in how the output is constructed, not just what is generated, allows diffusion models to cater to different preferences and task requirements, making them versatile tools for a broad spectrum of generation scenarios. We provide dynamic demonstrations of these capabilities at <https://hkunlp.github.io/blog/2025/dream/>.

这种基于时间步长的方法引入了推理时间缩放的额外维度（Snell 等人，2024；Muennighoff 等人，2025；Geiping 等人，2025），它与现有技术一起工作，例如 OpenAI o1（Jaech 等人，2024）和 Deep Seek R1（Guo 等人，2025）等大型语言模型中使用的思想链推理。这种可调整的计算质量权衡代表了扩散模型与传统自回归模型的区别的关键优势。

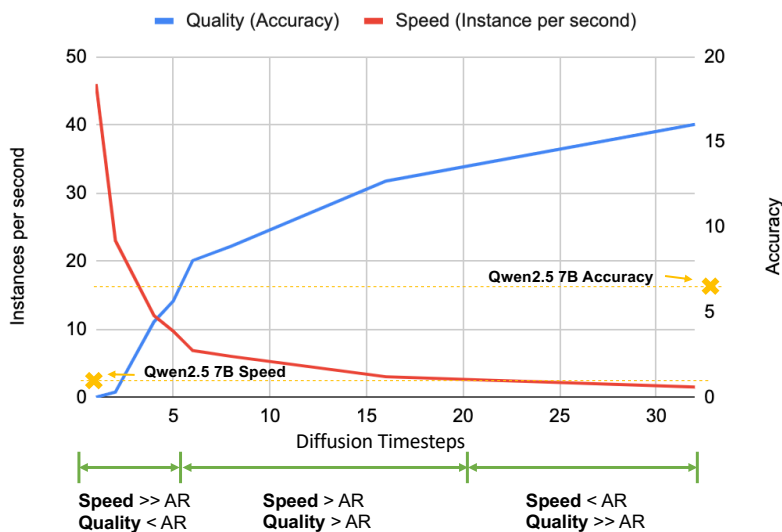


图 7: Dream 7B 和 Qwen2.5 7B 倒计时任务的质量速度比较。通过调整扩散时间步长，Dream 的性能可以灵活调整速度或质量。

5.5.3 任意订单生成

扩散模型表现出显著的推理灵活性，这使得它们与传统的自回归（AR）模型不同，后者通常以固定的从左到右的顺序生成序列。这种灵活性源于它们以任意顺序合成输出的固有能力，从而开辟了更广泛的多样化用户查询和交互模式。

Dream 7B 支持自然完成和填充任务，无需专门培训。这种灵活性的主要例子包括：

- 完成：用户可以提供一段初始文本，模型可以无缝地继续或完成它。
- 填充：扩散模型可以填充给定文本中缺失的片段。这可以不受特定要求的限制或指导，例如确保生成的填充导致精确的结束句子。
- 可配置的解码顺序：除了这些特定于任务的灵活性之外，用户还可以对解码行为本身进行控制。通过调整解码超参数，可以定制生成过程以模仿 AR 模型更结构化、从左到右的生成特征，或者相反，它可以在解码顺序中引入不同程度的随机性，从而朝着部分甚至完全随机顺序合成的方向发展。

这种输出构建方式（而不仅仅是生成内容）的适应性，使扩散模型能够满足不同的偏好和任务要求，使其成为适用于广泛生成场景的多功能工具。我们在 <https://hkunlp.github.io/blog/2025/dream/> 上提供这些功能的动态演示。

6 Conclusion

Dream 7B represents a significant step forward in diffusion-based language modeling, achieving competitive performance with state-of-the-art autoregressive models while providing unique capabilities for flexible text generation. Our experiments reveal that diffusion language models excel particularly in constraint-satisfaction and planning tasks, where their bidirectional nature and iterative refinement process offer clear advantages over traditional left-to-right generation approaches.

The model’s ability to perform arbitrary-order generation enables novel applications in completion and infilling tasks, while the adjustable timestep configurations provide users with fine-grained control over the quality-speed trade-off during inference. These capabilities, combined with the demonstrated effectiveness of initializing from pre-trained autoregressive models, establish a practical pathway for developing more flexible and capable language models. Future work will explore advanced post-training recipes, longer context capabilities, and applications to specialized domains requiring enhanced planning and reasoning abilities.

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6 结论

Dream 7B 代表了基于扩散的语言建模向前迈出的重要一步，实现了与最先进的自回归模型的竞争性能，同时为灵活的文本生成提供了独特的功能。我们的实验表明，扩散语言模型尤其在约束满足和规划任务中表现出色，其双向性质和迭代细化过程比传统的从左到右生成方法具有明显的优势。

该模型执行任意顺序生成的能力使得完成和填充任务中的新颖应用成为可能，而可调节的时间步长配置为用户提供了对推理期间质量与速度权衡的细粒度控制。这些功能与预先训练的自回归模型初始化的已证明的有效性相结合，为开发更灵活和更强大的语言模型建立了一条实用途径。未来的工作将探索先进的训练后方法、更长的上下文功能以及在需要增强规划和推理能力的专业领域的应用。

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